

DD-Mamba: A Dual-Domain State-Space Forecasting Framework with Linear Anchors and Zero-Initialized Corrections

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Abstract. Long-term multivariate forecasting must combine local time-domain dynamics with global periodic structure compact in the frequency domain. Mamba forecasters model one view but not both, and deep models often discard the simple linear maps that remain near state-of-the-art. We present DD-Mamba, a dual-domain *framework* whose time branch (decomposition with a selective state-space encoder) and frequency branch (learned spectral filtering) each produce a forecast, fused by a convex gate. Two principles organize it: *linear anchoring* — each branch carries a DLinear/FITS-style linear pathway with zero-initialized nonlinear heads, so at initialization the model is a linear map of the instance-normalized window; and *capacity placement* — a bidirectional Mamba over the *variate* dimension is switched on per dataset. We further separate normalization from dispersion *modeling*: RevIN is a reversible window-level normalization, not a scale forecast, so an optional dispersion head predicts a positive per-horizon scale from multi-resolution history, generalizing de-normalization toward explicit time-varying scale. On nine benchmarks under the look-back-96 protocol (three seeds), DD-Mamba reports the lowest mean MSE on seven, improving over reported S-Mamba on five (up to 17% on Solar) and tying three. Paired ablations validate capacity placement in both directions (the variate mixer helps weather yet hurts ETTh1; normalization helps weather yet hurt solar); data diagnostics show the datasets’ scale is non-stationary yet predictable, motivating the dispersion head. Dual-domain modeling is a per-dataset structural option, not a universal gain.

Keywords: Time series forecasting · State-space models · Mamba · Frequency domain · Multivariate analysis

1 Introduction

Long-term forecasting of multivariate time series underpins applications from power-grid load planning to traffic management and weather-dependent operations. The field has cycled through architectural paradigms at a remarkable pace: Transformer variants with efficient attention [23, 17, 24], patch-based attention [14], inverted variate-token attention [11], and, most recently, selective state-space models (SSMs) built on Mamba [7, 16], which offer linear-time sequence modeling with input-dependent dynamics.

Three empirical gaps complicate this progress. **Gap 1: deep models discard strong linear maps.** A per-component linear projection from the look-back window to the horizon (DLinear [21]) and a tiny complex-linear interpolation of the low-passed spectrum (FITS [19]) match or beat far larger models on several benchmarks, especially the ETT family; deep architectures that ignore this spend their early training re-discovering a linear map, and on small datasets often never do. **Gap 2: the useful inductive bias differs by domain.** Local trends and regime changes are naturally expressed in the time domain, while multi-scale seasonality is compact in the frequency domain, yet models committed to a single view — attention or SSMS over time steps only, or spectral mixing only [20] — systematically underuse the other. **Gap 3: the right amount of capacity is dataset-dependent.** Cross-variate mixing is the decisive component on high-dimensional benchmarks [11, 16] yet pure overfitting capacity on low-channel data, so any *fixed* architecture is mis-sized for one regime or the other.

We answer these three gaps with DD-Mamba (Fig. 1), mapping each to one structural remedy: *(i)* an explicit linear anchor with zero-initialized nonlinear corrections in each branch; *(ii)* parallel time and frequency branches, fused by a convex gate that preserves both prediction paths; and *(iii)* a bidirectional variate mixer switched on or off per dataset. Each remedy is detailed below.

Concretely: *(i) linear anchors* — each branch carries a DLinear-style [21] or FITS-style [19] linear pathway with zero-initialized nonlinear corrections, so at initialization the model reduces to a linear map of the instance-normalized window (the RLinear-class structure [9, 8]), a controlled test finding this accuracy-neutral; *(ii) dual branches* whose forecasts form a learned convex combination gated on both features, so each domain keeps an explicit path (ablations find the frequency branch significant on Solar and within noise elsewhere — per-dataset, not universal); and *(iii) capacity placement* — a causal Mamba over time and a bidirectional Mamba over variates, switched per dataset (removing the mixer hurts weather while adding it hurts ETTh1, each significant). Beyond these, an optional *dispersion head* generalizes RevIN de-normalization to *forecast* a positive per-horizon scale (Sect. 3.6), moving normalization from preprocessing toward explicit time-varying-scale modeling.

Contributions. (1) A dual-domain architecture whose output is a gated convex combination of per-branch forecasts; (2) a linear-anchoring scheme reducing to a linear map at initialization (accuracy-neutral in a controlled test); (3) a capacity-placement recipe for selective SSMS; (4) a dispersion head that forecasts a per-horizon scale, with data diagnostics showing the scale is non-stationary yet predictable; and (5) the lowest reported average MSE on seven of nine benchmarks (three-seed mean $\pm$ std), with a released harness and paired ablations including negative, dataset-dependent findings.

2 Related Work

Transformer forecasters. Informer [23], Autoformer [17] and FEDformer [24] adapt attention to long sequences via sparsity, decomposition, and frequency mixing; Non-stationary Transformers [12] restore the stationarity that normalization removes, and Crossformer [22] attends across both time and variates. PatchTST [14] treats sub-series patches as tokens with channel independence; iTransformer [11] inverts the axes and attends over *variates*, establishing that cross-channel modeling drives accuracy on high-dimensional benchmarks. DD-Mamba adopts the variate-mixing insight with linear-time bidirectional SSMs, coupled to an explicit frequency branch and linear anchors.

Linear and frequency-domain models. DLinear/NLinear [21] showed decomposition plus a linear map rivals Transformers; RLinear [9] added reversible instance normalization [8], and the MLP-based TiDE [4] and TimeMixer [15] confirm that lightweight mixers stay competitive. In the spectrum, FreTS [20] learns frequency-domain MLPs and FITS [19] forecasts with a single complex-linear interpolation of the low-passed spectrum at negligible cost. DD-Mamba does not compete with these models — it *contains* them: each branch carries a linear pathway of its domain’s family, with the deep machinery trained as a correction.

State-space models. Mamba [7] introduced selective SSMs with input-dependent dynamics and linear-time scans, extended by Mamba-2 [3]. In forecasting, S-Mamba [16] and TimeMachine [1] apply bidirectional Mamba over variate tokens, and further works explore frequency-domain SSMs [13]. DD-Mamba unifies these threads: causal Mamba over time, bidirectional Mamba over frequency bins and variates, placed per dataset rather than uniformly.

Decomposition and heteroscedasticity. Splitting a series into interpretable components is a century-old idea: classical additive models, STL [2], wavelet analysis, EMD, and SSA extract a trend, a seasonal part, and an irregular remainder, and learned decompositions such as SCINet [10] carry it into deep forecasters. A recurring limitation, made explicit by the seasonal-trend-*dispersion* decomposition (STD) [5], is that these methods model *level* but not *scale*: when a series’ variance changes over time — heteroscedasticity — the fluctuation is left in the irregular term. We relate to this lineage at two clearly separated levels. *Diagnosis:* RevIN [8] performs reversible, window-level normalization by the *input* mean and standard deviation; it is emphatically *not* a dispersion decomposition and does not *forecast* future scale, but by removing per-window scale it exposes whether a dataset’s apparent nonstationarity is genuine scale shift (which it absorbs) or periodic structure (which it distorts) — a distinction Sect. 3.6 measures directly. *Modeling:* an optional dispersion head (Sect. 3.6) predicts a positive per-horizon, per-variate scale $\hat{\mathbf{D}}$ from multi-resolution history and reconstructs $\hat{\mathbf{Y}} = \hat{\mathbf{T}} + \hat{\mathbf{D}} \odot \hat{\mathbf{S}}$, a genuine “trend plus dispersion times seasonal shape” in the spirit of STD — though we *predict* scale rather than *extract* it, and make no claim of producing an STD/STDR remainder. The seasonal-trend split itself

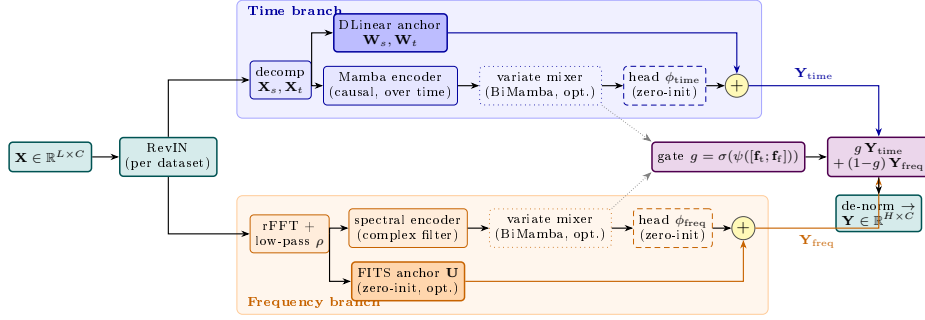


Fig. 1. DD-Mamba overview, color-coded by role: the **normalization** wrapper (teal), the **time branch** (blue), the **frequency branch** (orange), and the **convex-gate fusion** (violet). In each branch a solid-filled *linear anchor* runs in parallel with a zero-initialized nonlinear head (dashed), so at initialization the model reduces to a linear map of the instance-normalized window. Dotted boxes are per-dataset switches; dotted grey arrows carry branch features to the gate, whose convex weight g mixes the two coloured forecast paths $\mathbf{Y}_{\text{time}}$ and $\mathbf{Y}_{\text{freq}}$.

lands on the *linear anchors* (Sect. 3.2). The same discipline of routing a specific deficiency to a dedicated operator has been argued for frequency-domain design in adjacent fields [6].

3 Method

3.1 Problem Formulation and Normalization

Given a look-back window $\mathbf{X} \in \mathbb{R}^{L \times C}$ of L steps and C variates, the task is to predict the next H steps $\mathbf{Y} \in \mathbb{R}^{H \times C}$. Following RevIN [8], we standardize each window per variate, $\tilde{\mathbf{X}} = (\mathbf{X} - \boldsymbol{\mu})/\boldsymbol{\sigma}$, and de-normalize the final forecast. Instance normalization is treated as a per-dataset switch rather than a universal default; Sect. 4.3 shows why.

3.2 Time Branch: Decomposition, Linear Backbone, Selective Scan

The time branch decomposes $\tilde{\mathbf{X}}$ with a moving average into trend $\mathbf{X}_t$ and seasonal $\mathbf{X}_s$ components [17, 21] and forecasts with two cooperating paths.

Linear backbone. Per-component linear maps project the window directly to the horizon:

$$\mathbf{Y}_{\text{lin}} = \mathbf{W}_s \mathbf{X}_s + \mathbf{W}_t \mathbf{X}_t \in \mathbb{R}^{H \times C}, \quad \mathbf{W}_s, \mathbf{W}_t \in \mathbb{R}^{H \times L}, \quad (1)$$

i.e., a DLinear-style map [21] (randomly initialized and trained jointly — a parameterized pathway, not a fitted DLinear), shared across variates.

Selective-scan correction. The pair $[\mathbf{X}_s; \mathbf{X}_t]$ is embedded per time step and processed channel-independently by a stack of Mamba layers [7] (Fig. 2a). A

Mamba layer discretizes a state-space system with input-dependent parameters $(\Delta, \mathbf{B}, \mathbf{C})$:

$$\mathbf{h}_k = e^{\Delta_k \mathbf{A}} \mathbf{h}_{k-1} + \Delta_k \mathbf{B}_k u_k, \quad y_k = \mathbf{C}_k^\top \mathbf{h}_k + D u_k, \quad (2)$$

scanned causally over the L steps and applied independently to each of the d_{in} feature dimensions: per dimension, the state is $\mathbf{h}_k \in \mathbb{R}^N$, $\mathbf{A} \in \mathbb{R}^N$ is diagonal, the input u_k is scalar, and $(\Delta_k \in \mathbb{R}, \mathbf{B}_k, \mathbf{C}_k \in \mathbb{R}^N)$ are produced from u by learned linear maps [7]. The last state, summed with a whole-window linear embedding $\mathbf{W}_e \tilde{\mathbf{x}}^{(e)}$ (one per variate, as in [16]), forms the branch feature $\mathbf{f}_{\text{time}} \in \mathbb{R}^{C \times d}$. A head ϕ_{time} maps features to a horizon correction:

$$\mathbf{Y}_{\text{time}} = \mathbf{Y}_{\text{lin}} + \phi_{\text{time}}(\mathbf{f}_{\text{time}}), \quad \phi_{\text{time}} \equiv \mathbf{0} \text{ at init.} \quad (3)$$

Because ϕ_{time} is zero-initialized, the branch’s output at initialization equals its (randomly initialized) DLinear-style backbone; the SSM learns only what the linear map cannot express. We run the recurrence in fp32 even under mixed precision, as the L -step accumulation overflows in half precision and silently destroys training.

3.3 Frequency Branch: Spectral Filtering and a FITS Anchor

The frequency branch applies the real FFT along time, $\mathbf{Z} = \text{rFFT}(\tilde{\mathbf{X}}) \in \mathbb{C}^{C \times F}$, $F = \lfloor L/2 \rfloor + 1$, optionally low-passed by zeroing the top fraction ρ of bins.

Spectral encoder. A learnable complex linear filter $\mathbf{W} = \mathbf{W}_r + i\mathbf{W}_i \in \mathbb{C}^{F \times F}$ densely mixes bins, $\mathbf{Z}' = \mathbf{Z}\mathbf{W}^\top$ along the frequency axis (a bidirectional Mamba over the bin sequence is an input-dependent alternative). The (real, imaginary) parts are projected to the branch feature $\mathbf{f}_{\text{freq}} \in \mathbb{R}^{C \times d}$ and a head ϕ_{freq} produces the spectral forecast.

FITS anchor (optional, per dataset). Mirroring the time branch, a complex linear map $\mathbf{U} \in \mathbb{C}^{F' \times F}$ with $F' = \lfloor (L+H)/2 \rfloor + 1$, applied along the frequency axis ($\mathbf{Z}\mathbf{U}^\top \in \mathbb{C}^{C \times F'}$), maps the (low-passed) window spectrum to that of a length- $(L+H)$ series, whose inverse transform’s last H samples are the branch’s linear spectral forecast:

$$\mathbf{Y}_{\text{freq}} = \underbrace{\text{irFFT}_{L+H}(\mathbf{Z}\mathbf{U}^\top)[L:]}_{\text{linear-in-frequency}} + \phi_{\text{freq}}(\mathbf{f}_{\text{freq}}), \quad \mathbf{U}, \phi_{\text{freq}} \equiv \mathbf{0} \text{ at init.} \quad (4)$$

This pathway *parameterizes* the FITS [19] family, but unlike FITS itself $\mathbf{U}$ is *zero-initialized*, so the frequency branch is silent at initialization (the length- L and length- $(L+H)$ frequency grids do not align, so no canonical identity init exists, and a random $\mathbf{U}$ would inject noise during the highest-learning-rate epochs). With the time head, spectral map, and gate all zero or constant at init, the end-to-end map in the normalized space is the linear $g_0 \cdot \mathbf{Y}_{\text{lin}}$; composed with RevIN this is the RLinear-class structure [9, 8] — linear in the normalized window, by construction not fitted. The anchor is enabled per dataset (Table 2); where disabled, linear anchoring rests on the time branch alone.

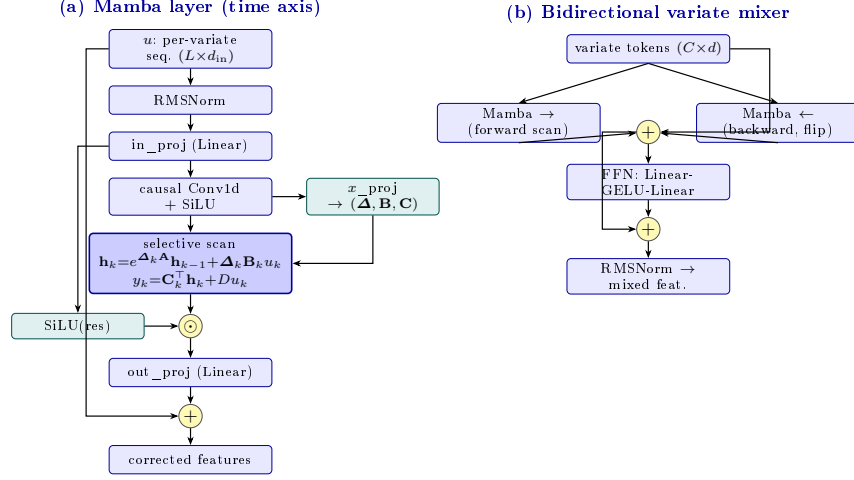


Fig. 2. Internal structure of DD-Mamba’s two selective-state-space blocks (cf. the blue boxes in Fig. 1). **(a)** The causal Mamba layer used as the time-axis encoder (Eq. (2)): an input-dependent selective scan, gated by a SiLU branch and wrapped in a residual, with $(\Delta, \mathbf{B}, \mathbf{C})$ predicted per step from the input. **(b)** The bidirectional mixer used over the *variate* dimension (Sect. 3.4): forward and backward scans of the C channel tokens summed in one residual stream, then a position-wise feed-forward block (the S-Mamba recipe [16]).

3.4 Cross-Channel Mixing over Variates

Channel independence is a strong baseline [14, 21] but leaves accuracy on the table when variates are coupled [11, 16]. Inside each branch, before its head, we optionally run M layers of *bidirectional* Mamba over the C per-channel features, followed by a position-wise feed-forward block (Fig. 2b). Bidirectionality is essential: channel order carries no causal direction, so each layer sums a forward and a reversed scan in one residual stream. The mixer is a per-dataset choice: $M=2$ at width 512 on electricity/traffic (its single most load-bearing component; Sect. 4.3), $M=2$ at width 256 on weather, $M=1$ on solar, and $M=0$ on the few-channel ETT and exchange datasets, where it triples parameters without adding signal.

3.5 Forecast-Level Convex Fusion

Each branch emits a forecast and a feature. The model output is a gated convex combination

$$\mathbf{Y} = g \odot \mathbf{Y}_{\text{time}} + (1 - g) \odot \mathbf{Y}_{\text{freq}}, \quad g = \sigma(\psi([\mathbf{f}_{\text{time}}; \mathbf{f}_{\text{freq}}])) \in (0, 1)^{C \times 1}, \quad (5)$$

with ψ applied row-wise (per variate), followed by RevIN de-normalization. Since $\mathbf{Y}$ lies elementwise between the branch forecasts, both domains retain an explicit

prediction path, and the gate value reports each branch’s convex mixing weight — a lightweight proxy for its contribution, not a causal attribution. The constraint is structural, not behavioral: g is a sigmoid and can saturate near 0 or 1, so one branch may contribute very little — by design, as the ablations show the useful domain is dataset-dependent. ψ is zero-initialized with bias 2.2, so training starts at $g \approx 0.9$: the time branch is linear at initialization while the frequency branch is silent, and an even 0.5/0.5 mix would waste the highest-learning-rate epochs on noise.

3.6 Dispersion Head: Forecasting Time-Varying Scale

RevIN reconstructs a forecast as $\mathbf{Y} = \boldsymbol{\mu} + \boldsymbol{\sigma} \odot \mathbf{S}$ with the input-window mean $\boldsymbol{\mu}$ and standard deviation $\boldsymbol{\sigma}$ held *constant* over the horizon — a reversible normalization, not a forecast of how the scale evolves. We optionally generalize the reconstruction to

$$\hat{\mathbf{Y}} = \hat{\mathbf{T}} + \hat{\mathbf{D}} \odot \mathbf{S}, \quad \hat{\mathbf{D}} = \boldsymbol{\sigma} \odot \text{softplus}(\psi_D(\mathbf{s})) \in \mathbb{R}_{>0}^{C \times H}, \quad \hat{\mathbf{T}} = \boldsymbol{\mu} + \psi_T(\mathbf{s}), \quad (6)$$

where $\mathbf{S}$ is the normalized-space fused forecast (the shape) and $\hat{\mathbf{D}}$ is a *positive, per-horizon, per-variate* scale — unlike RevIN’s single $\boldsymbol{\sigma}$ per variate. The heads ψ_D, ψ_T are zero-initialized so that $\hat{\mathbf{D}} = \boldsymbol{\sigma}$ and $\hat{\mathbf{T}} = \boldsymbol{\mu}$ at start: the module *begins* exactly as RevIN de-normalization and learns per-horizon corrections, keeping $\hat{\mathbf{D}}$ anchored to the input scale and strictly positive. This is STD’s “trend plus dispersion times seasonal shape” with the scale *predicted*, not extracted. *Multi-resolution history*: the heads read $\mathbf{s}$, the per-variate mean and std of the past $\{96, 144, 288, 336\}$ samples ending at the forecast origin (only pre-origin information), so the scale predictor sees cycles longer than the $L=96$ look-back — Solar’s 144-step daily and Traffic’s weekly periods — without enlarging the branch inputs. A “fixed” variant instead sets $\hat{\mathbf{D}}$ to the longest-resolution historical std (non-learned).

3.7 Training

We minimize MSE with AdamW under a halving schedule ($\eta_e = \eta_0 \cdot 2^{-(e-1)}$), which concentrates learning in the first epochs and suppresses the late-epoch overfitting that cosine schedules invite on small benchmarks; all runs use 10 epochs with best-checkpoint early stopping.

4 Experiments

4.1 Setup

Datasets and protocol. We evaluate on nine standard benchmarks (Table 1) under the unified protocol of [11, 16]: look-back 96, horizons 96/192/336/720. ETT uses the canonical 12/4/4-month border split, the others chronological 0.7/0.1/0.2 splits; variates are standardized by training-set statistics; metrics are MSE and MAE.

Table 1. Benchmark datasets and where DD-Mamba places its capacity.

Dataset	Variates	Steps	Granularity	Variate mixer	Params
ETTh1/ETTh2	7	17,420	1 hour	off	0.25M
ETTm1/ETTm2	7	69,680	15 min	off	0.24–0.36M
Weather	21	52,696	10 min	$M=2$, $d=256$	5.8M
Solar-Energy	137	52,560	10 min	$M=1$, $d=128$	0.96M
Electricity	321	26,304	1 hour	$M=2$, $d=512$	19.7M
Traffic	862	17,544	1 hour	$M=2$, $d=512$	19.7M
Exchange	8	7,588	1 day	off	0.09M

Baselines. We compare against the strongest published results under this protocol: S-Mamba [16], iTransformer [11], PatchTST [14], DLinear [21], and TimesNet [18]. Numbers are quoted from [16] rather than reproduced: the protocol is identical, but training pipelines, seeds, and implementations differ across works, so sub-noise-floor gaps to quoted numbers should be read with caution (flagged in the text); a unified rerun is planned for the camera-ready appendix.

Implementation and Reproducibility. All experiments use PyTorch 2.11 (CUDA 13.0) on a single NVIDIA RTX 5090. Optimization is AdamW (grad clip 5.0), 10 epochs with the learning rate halved each epoch, early stopping on validation loss, and mixed precision with the selective scan forced to fp32 (Sect. 3.2). The decomposition kernel is 25; Mamba blocks use expansion 2 and conv width 4 throughout. Table 2 lists all per-dataset settings; configs and the ablation harness ship with the code. All main results are means over *three seeds* (2024–2026); the maximum per-entry std across the 36 dataset–horizon entries is 0.005 MSE, typically 0.001–0.003.

Statistical methodology. We compare at two levels. *Within our framework* (ablations), every variant shares seeds with the reference, so we report *paired* per-seed differences with t -based 95% CIs ($n=3$) and treat a component as supported only when its CI excludes zero. *Against quoted baselines*, pairing is impossible, so we use a conservative unpaired tie threshold of 0.005 MSE. Our in-framework DLinear re-run shows why: through the identical pipeline on ETTh1 it averages 0.462/0.456 MSE/MAE versus its quoted 0.456/0.452 — a +0.006 implementation gap, the order of several margins in Table 3. Cross-paper comparisons therefore cannot support sub-0.01 distinctions; DD-Mamba’s margin over the in-framework DLinear (0.439 vs. 0.462) is, by contrast, a same-pipeline comparison.

4.2 Main Results

Under the noise-floor tie rule, DD-Mamba reports the lowest average MSE on seven of nine datasets (Table 3; per-horizon values in Table 4). The margin is

Table 2. Per-dataset hyperparameters. d : model width; time enc.: time-branch encoder (Mamba layers in parentheses); M : variate-mixer layers; N : SSM state size; ρ : spectral low-pass fraction (Sect. 3.3); FITS: the *optional* spectral linear anchor of Eq. (4) — when “no”, the frequency branch remains fully active (rFFT, spectral encoder, forecast head) but carries no explicit linear pathway; IN: instance normalization (RevIN); do.: dropout; B: batch size; wd: weight decay. [†]Solar adopts IN = no from the ablation finding (Sect. 4.3); its main results were re-measured under this configuration across all horizons and seeds.

Dataset	d	time enc.	M	N	ρ	FITS	IN	lr	B	do./wd
ETTh1	128	Mamba (1)	0	16	0.4	yes	yes	$5 \cdot 10^{-4}$	32	$0.2/10^{-4}$
ETTh2	128	Mamba (1)	0	16	0.3	yes	yes	$5 \cdot 10^{-4}$	32	$0.3/5 \cdot 10^{-4}$
ETTh1	128	Mamba (2)	0	16	0	no	yes	10^{-4}	32	$0.1/10^{-4}$
ETTh2	128	Mamba (1)	0	16	0.2	no	yes	10^{-4}	32	$0.2/2 \cdot 10^{-4}$
Weather	256	Mamba (2)	2	16	0	yes	yes	$2 \cdot 10^{-4}$	32	$0.1/10^{-4}$
Solar	128	Mamba (2)	1	16	0	no	no [†]	$5 \cdot 10^{-4}$	16	$0.1/10^{-4}$
Electricity	512	MLP	2	32	0	no	yes	$5 \cdot 10^{-4}$	16	$0.1/10^{-4}$
Traffic	512	MLP	2	32	0	no	yes	10^{-3}	16	$0.1/10^{-4}$
Exchange	64	Mamba (1)	0	8	0.5	no	yes	10^{-4}	32	$0.3/5 \cdot 10^{-4}$

Table 3. Long-term forecasting results averaged over horizons $H \in \{96, 192, 336, 720\}$ with look-back $L=96$ (MSE/MAE; lower is better). DD-Mamba entries are means over three seeds (Table 4 reports per-horizon std). Baselines quoted from [16]. **Bold** marks the best result *and* every result within the run-to-run noise floor (< 0.005 , Sect. 4.1) of it, treated as tied best; underline marks the next-best result.

Dataset	DD-Mamba (ours)	S-Mamba	iTransformer	PatchTST	DLinear	TimesNet
ETTh1	0.439/0.434	0.455/0.450	<u>0.454/0.447</u>	0.469/0.454	0.456/0.452	0.458/0.450
ETTh2	0.374/0.401	<u>0.381/0.405</u>	0.383/ <u>0.407</u>	0.387/ <u>0.407</u>	0.559/0.515	0.414/0.427
ETTh1	0.386/0.398	<u>0.398/0.405</u>	0.407/0.410	0.387/0.400	0.403/0.407	0.400/0.406
ETTh2	0.280/0.325	<u>0.288/0.332</u>	<u>0.288/0.332</u>	0.281/0.326	0.350/0.401	0.291/0.333
Weather	0.248/0.276	0.251/0.276	<u>0.258/0.278</u>	0.259/ <u>0.281</u>	0.265/0.317	0.259/0.287
Electricity	0.168/0.263	0.170/0.265	<u>0.178/0.270</u>	0.205/0.290	0.212/0.300	0.192/0.295
Traffic	<u>0.438/0.281</u>	0.414/0.276	<u>0.428/0.282</u>	0.481/0.304	0.625/0.383	0.620/0.336
Exchange	0.363/ 0.405	0.367/ <u>0.408</u>	<u>0.360/0.403</u>	0.367/ 0.404	0.354/0.414	0.416/0.443
Solar	0.199/0.260	0.240/ <u>0.273</u>	<u>0.233/0.262</u>	0.270/0.307	0.330/0.401	0.301/0.319

clear on Solar (0.199 vs. iTransformer’s 0.233; -17% vs. S-Mamba, with the normalization-free configuration of Sect. 4.3), ETTh1 (-0.015), and ETTh2 (-0.007); on ETTh1/ETTh2 it beats S-Mamba but ties PatchTST, and on Weather/Electricity the lead is below the floor (tied). It stays competitive on Exchange (a near random-walk where DLinear leads by 0.009) and trails S-Mamba on Traffic by $+0.024$, which we *hypothesize* reflects the weekly cycle exceeding the look-back. As all comparisons are to numbers *reported* in [16], we phrase the claim conservatively: DD-Mamba is competitive with results reported under this protocol, its largest margins on Solar (normalization) and the small ETT datasets, where the anchors and capacity restraint matter most and DD-Mamba uses $\approx 0.25M$ parameters — two orders of magnitude fewer than variate-attention models.

Table 4. DD-Mamba per-horizon results (mean $\pm$ std over three seeds), look-back $L=96$.

Dataset		$H=96$	$H=192$	$H=336$	$H=720$	Avg
ETTh1	MSE	0.380 $\pm$.001	0.431 $\pm$.002	0.474 $\pm$.004	0.470 $\pm$.003	0.439
	MAE	0.396 $\pm$.001	0.424 $\pm$.001	0.447 $\pm$.003	0.468 $\pm$.001	0.434
ETTh2	MSE	0.293 $\pm$.001	0.368 $\pm$.003	0.414 $\pm$.001	0.421 $\pm$.001	0.374
	MAE	0.344 $\pm$.002	0.391 $\pm$.000	0.426 $\pm$.000	0.441 $\pm$.001	0.401
ETTM1	MSE	0.320 $\pm$.001	0.366 $\pm$.002	0.398 $\pm$.001	0.462 $\pm$.001	0.386
	MAE	0.359 $\pm$.001	0.384 $\pm$.001	0.406 $\pm$.001	0.443 $\pm$.001	0.398
ETTM2	MSE	0.177 $\pm$.000	0.242 $\pm$.002	0.302 $\pm$.000	0.401 $\pm$.003	0.280
	MAE	0.260 $\pm$.001	0.303 $\pm$.001	0.340 $\pm$.000	0.398 $\pm$.002	0.325
Weather	MSE	0.162 $\pm$.000	0.212 $\pm$.000	0.269 $\pm$.001	0.352 $\pm$.003	0.248
	MAE	0.207 $\pm$.001	0.255 $\pm$.000	0.295 $\pm$.002	0.349 $\pm$.002	0.276
Electricity	MSE	0.140 $\pm$.000	0.157 $\pm$.001	0.174 $\pm$.001	0.200 $\pm$.004	0.168
	MAE	0.235 $\pm$.000	0.252 $\pm$.001	0.271 $\pm$.001	0.296 $\pm$.004	0.263
Traffic	MSE	0.402 $\pm$.002	0.424 $\pm$.001	0.441 $\pm$.001	0.485 $\pm$.003	0.438
	MAE	0.266 $\pm$.001	0.275 $\pm$.001	0.283 $\pm$.000	0.301 $\pm$.000	0.281
Exchange	MSE	0.086 $\pm$.000	0.179 $\pm$.001	0.330 $\pm$.002	0.857 $\pm$.001	0.363
	MAE	0.205 $\pm$.000	0.301 $\pm$.000	0.416 $\pm$.002	0.699 $\pm$.000	0.405
Solar	MSE	0.180 $\pm$.004	0.195 $\pm$.005	0.207 $\pm$.001	0.212 $\pm$.005	0.199
	MAE	0.237 $\pm$.003	0.259 $\pm$.003	0.270 $\pm$.001	0.274 $\pm$.005	0.260

4.3 Component Ablations

Our ablation harness flips exactly one switch of a dataset’s full configuration, holding split, schedule, seeds, and every other hyperparameter fixed. We report its complete output on three datasets under their *final released configurations*, three seeds each at $H=96$: ETTh1 and Solar (Table 5) and Weather (Table 6), with the paired methodology of Sect. 4.1. Two consistency checks pass: the Solar “full” row reproduces the main-table Solar $H=96$ entry (0.180 $\pm$.004), and its w/o-instance-norm variant coincides with the full (RevIN-off) configuration, so that cell is omitted.

Dual-domain evidence. The decisive result is on Solar: removing the frequency branch degrades MSE by +0.0114 with a CI excluding zero — the first statistically supported frequency-branch contribution in our studies. On ETTh1 and Weather both single-branch variants lie within their CIs (on ETTh1 the frequency branch *alone* even matches the full model, -0.0007): the branches are largely redundant there. The dual-domain claim is thus conditional — the frequency path contributes measurably on Solar, redundant but not harmful elsewhere.

Capacity placement, validated both ways. Adding the variate mixer on ETTh1 *hurts* (+0.0077*) while removing it on Weather also hurts (+0.0104*); normalization shows the same two-sided pattern (removing it hurts Weather +0.0161*, while disabling it on Solar improved the re-measured main results from 0.228 to 0.199). Zero initialization is a *null* result — no significant difference on either dataset (-0.0012 ETTh1, -0.0046 Solar; CIs include zero) — so we make no accuracy claim for it, only the interpretable linear start (Sect. 3.2). The DLinear backbone and the learned gate versus averaging are within CIs everywhere; on

Table 5. Component ablations on ETTh1 and Solar-Energy ($H=96$, three seeds, final configurations): paired per-seed Δ MSE to the full model with t -based 95% CIs; * CI excludes zero. The FITS anchor is enabled on ETTh1 (hence *removed*) and disabled on Solar (hence *added*); the variate mixer is off on ETTh1 (*added*) and on for Solar (*removed*).

Variant	ETTh1: Δ [95% CI]	Solar: Δ [95% CI]
full (reference MSE)	0.3802 $\pm$.001	0.1802 $\pm$.004
time branch only (freq removed)	+0.0034 [−0.009, +0.016]	+0.0114 [+0.001, +0.021]*
freq branch only (time removed)	−0.0007 [−0.004, +0.003]	+0.0051 [−0.013, +0.023]
sum fusion	−0.0026 [−0.007, +0.002]	+0.0066 [−0.005, +0.018]
w/o instance norm	+0.0061 [−0.001, +0.013]	—
w/o linear backbone	−0.0031 [−0.009, +0.003]	+0.0081 [−0.019, +0.035]
random init (vs. zero-init)	−0.0012 [−0.005, +0.002]	−0.0046 [−0.022, +0.013]
FITS anchor removed / added	+0.0042 [−0.018, +0.026]	+0.0088 [−0.002, +0.019]
variate mixer added / removed	+0.0077 [+0.004, +0.012]*	+0.0119 [−0.010, +0.034]
Mamba→MLP over time	−0.0022 [−0.007, +0.003]	+0.0089 [−0.002, +0.020]
freq Mamba encoder	−0.0024 [−0.011, +0.006]	−0.0117 [−0.044, +0.020]

Table 6. Component ablation on Weather ($H=96$; MSE mean $\pm$ std over three seeds). Δ is the *paired* per-seed difference to the full model with a t -based 95% confidence interval ($n=3$); * CI excludes zero.

Variant	MSE	paired Δ [95% CI]	Component isolated
full	0.1620 $\pm$.0003	—	reference
w/o instance norm	0.1781 $\pm$.0035	+0.0161 [+0.0053, +0.0269]*	RevIN
w/o channel mixer	0.1724 $\pm$.0021	+0.0104 [+0.0037, +0.0172]*	cross-variate Mamba mixing
Mamba→MLP over time	0.1647 $\pm$.0005	+0.0027 [+0.0020, +0.0034]*	selective scan (time)
w/o linear backbone	0.1631 $\pm$.0006	+0.0011 [−0.0014, +0.0036]	DLinear anchor (time)
time branch only	0.1628 $\pm$.0011	+0.0007 [−0.0020, +0.0035]	frequency branch removed
w/o FITS anchor	0.1627 $\pm$.0016	+0.0007 [−0.0041, +0.0055]	spectral linear anchor
freq branch only	0.1626 $\pm$.0014	+0.0006 [−0.0040, +0.0052]	time branch removed
freq Mamba encoder	0.1626 $\pm$.0016	+0.0006 [−0.0044, +0.0055]	spectral encoder type
sum fusion	0.1622 $\pm$.0007	+0.0001 [−0.0020, +0.0023]	learned gate → average

Weather (Table 6) exactly three components survive the paired test (normalization, the variate mixer, and the temporal Mamba, +0.0027, positive on all seeds). The recurring pattern — component value is dataset-dependent, in *both* directions — is the argument for per-dataset switches over universal defaults.

Dispersion diagnostics: is the time-varying scale real? We test the interpretation directly (Fig. 3; no model training). Standardizing each channel and taking a rolling std over the dominant period, all three probed datasets carry substantial time-varying scale (coefficient of variation 0.20–0.48); a KPSS test rejects stationarity of the rolling-std series in *every* sampled channel, and a Ljung-Box test finds it strongly autocorrelated — the scale is non-stationary and *predictable*, exactly what a dispersion head can exploit. Yet the benefit of *normalizing* by it is dataset-dependent: removing RevIN helps solar (−0.029 avg MSE) but hurts traffic (+0.100). The distinction is structural, not statistical — solar’s night zeros are *periodic structure* (window statistics encode the night fraction, and de-normalization amplifies errors by an unstable std), whereas traffic windows carry genuine distribution shift (weekday/weekend, per-sensor levels) that nor-

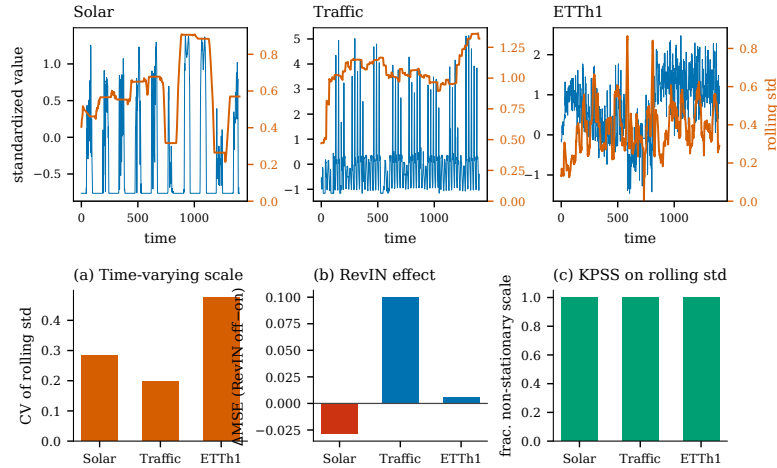


Fig. 3. Dispersion diagnostics (no training). *Top*: a representative standardized channel (blue) with its rolling standard deviation (orange) for Solar, Traffic, and ETTh1 — the local scale visibly varies over time (Solar’s is strongly periodic, tracking the day/night cycle). **(a)** The coefficient of variation of the rolling std quantifies this time-varying scale. **(b)** The measured RevIN effect (ΔMSE , off-on): normalizing by the input scale *helps* traffic but *hurts* solar. **(c)** A KPSS test rejects stationarity of the rolling-std series in every sampled channel — the scale itself is non-stationary — motivating a head that forecasts it.

malization absorbs. We therefore ship normalization as a verified per-dataset switch, and position the dispersion head (Sect. 3.6) as the route to *forecast* the predictable component of the scale rather than only normalize by its input estimate.

Dispersion head: evaluation protocol. We evaluate the head with a four-variant chain under shared seeds, splits, and budget — base (no normalization), then RevIN, fixed historical dispersion, and the learned head (Eq. (6)) — reporting, besides MSE/MAE, the error on high- vs. low-variability windows and, to match the interpretability standard of [5], ADF/KPSS/Ljung-Box statistics of the residual *before and after* the head (does explicit scale prediction remove heteroscedasticity that would otherwise stay in the error?). The head, multi-resolution statistics, and this chain ship in the released harness; a same-pipeline probe confirms it trains stably and reduces to RevIN at initialization. As the diagnostics predict a dataset-dependent outcome, a head that helps only high-variability datasets would still support the per-dataset placement claim. The full chain with the residual battery, on the RTX 5090, is reported in the extended version.

Negative results that shaped the design. Deepening the traffic mixer from two to four layers ($19.7 \rightarrow 37.9\text{M}$) gave no gain, so the release keeps two; enabling the

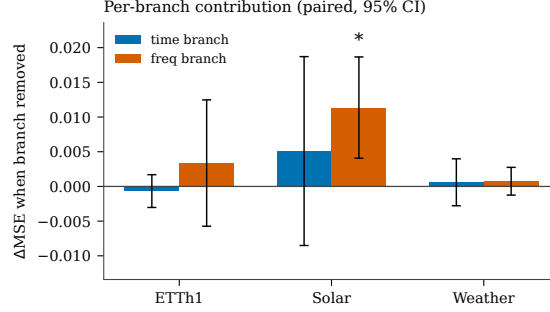


Fig. 4. Per-branch contribution: the paired increase in MSE ($H=96$, three seeds) when each branch is removed — freq contribution = time-only — full, time contribution = freq-only — full — with t -based 95% CIs; * marks a CI clearing zero. Only Solar’s frequency branch is individually significant; on ETTh1 and Weather the branches are mutually redundant — the conditional dual-domain claim made visual.

mixer on ETT (8.4K training windows) tripled parameters and *caused* premature early stopping through overfitting, so the released ETT configurations disable it.

4.4 Analysis

Where each domain wins. On solar the branch-only ablations (Table 5, Fig. 4) show both domains contribute but the time view carries more (removing it costs +0.012 vs. +0.002 MSE) — consistent with the look-back hiding periodic structure (the 10-minute daily cycle spans $144 > L=96$ samples). The fusion gate (Eq. (5)) exposes per-sample convex weights as a lightweight proxy: on a trained ETTh1 model the gate settles below equal mix, agreeing with the ablation (though only the ablation is causal).

Efficiency profile. Table 7 reports per-dataset measurements that are hardware-independent. Parameter count is governed by model *width*, not channel count (electricity and traffic share 19.7M despite a $2.7\times$ gap in variates), so model size is decoupled from data dimensionality; compute and activation memory instead scale with the channel count through the variate mixer (traffic 60 GMACs vs. electricity’s 22 at equal width, while ETT and exchange sit two orders of magnitude below), making the capacity restraint there cheap as well as accurate. Wall-clock figures are left for the extended version.

5 Conclusion

DD-Mamba is a dual-domain state-space forecasting *framework*: each branch carries a linear pathway, deep components are zero-initialized corrections, and

Table 7. Engineering profile of DD-Mamba per dataset (look-back 96, horizon 96, batch 1). GMACs: multiply-accumulates per forecast (**Linear** + **Conv1d** + the selective-scan recurrence); act. mem.: forward activation footprint (fp32). All three columns are hardware-independent; wall-clock throughput on the RTX 5090 is deferred to the extended version.

Dataset	Variates	Params (M)	GMACs	Act. mem. (MB)
Exchange	8	0.09	0.02	2.3
ETTh1	7	0.25	0.08	4.0
Weather	21	5.77	1.90	46.6
Solar	137	0.96	3.26	150.4
Electricity	321	19.70	22.4	254.3
Traffic	862	19.70	60.2	682.7

capacity (especially cross-variate mixing) is placed per dataset. Under the look-back-96 protocol it achieves the lowest reported average MSE on seven of nine benchmarks at 0.09–19.7M parameters. The paired ablations validate capacity placement in *both* directions (the variate mixer helps weather yet hurts ETTh1; normalization helps weather yet hurt solar), find the frequency branch significant on Solar, and find zero initialization accuracy-neutral. Separately, we sharpen the relation to dispersion decomposition: RevIN normalizes but does not *forecast* scale, so an optional dispersion head predicts a per-horizon scale from multi-resolution history — with diagnostics confirming the scale is non-stationary yet predictable. We thus claim per-dataset structural flexibility, not a universal gain, plus a caution: normalization helps or hurts depending on whether apparent nonstationarity is genuine shift or periodic structure.

Limitations, each with its resolving experiment: (1) the frequency-branch gain is shown on one dataset ($n=3$); (2) zero initialization’s stability value is untested; (3) baselines are quoted from [16], not re-run (the +0.006 DLinear gap, Sect. 4.1); and (4) the dispersion head’s full four-variant chain across datasets, with the residual-stationarity battery, is reported in the extended version. The fixed 96-step look-back hides long cycles (traffic’s weekly) that the multi-resolution dispersion statistics begin to address.

Reproducibility. Code, configurations, and the ablation harness are supplementary material, to be released publicly upon acceptance.

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